

DA3408 AI Operations — Module 1 Assignment

Experiment Management & Reproducibility with MLflow + DVC · Samrudh · <https://github.com/samrudh123/DA3408-Assignment-1>

Q1 — Technical Debt Diagnosis (written; full answer in q1/aiops_q1.pdf)

(a) Rounding change to *estimated delivery time* degrading *favourite restaurants* $\Rightarrow$ **boundary erosion / entanglement**, the CACE principle: weights are learned jointly, so shifting one feature's distribution re-weights every other feature on retrain. The tell is that the effect was far larger than the code change.

(b) An unknown marketing team reading the model's output table $\Rightarrow$ **undeclared consumers (visibility debt)**: with no declared interface the ML team cannot cost any schema or model change, and the campaign it drives risks closing a hidden feedback loop back into the training data.

(c) 14 undocumented shell scripts with no orchestration $\Rightarrow$ **pipeline jungle**. *Mitigation*: replace them with a `dvc.yaml` DAG (ingest $\rightarrow$ clean $\rightarrow$ featurize $\rightarrow$ train $\rightarrow$ evaluate), rebuild with `dvc repro` so only changed stages re-run, version data via `dvc add/dvc push`, lift hyperparameters into `params.yaml`, and log every run to MLflow with a `git_commit` tag.

Q2 — MLflow Experiment Comparison (q2/Q2_Mlflow.ipynb)

Six runs of an `MLPClassifier` on MNIST (12k subset, 30 epochs) over a 3×2 grid: `hidden_layer_sizes` $\in \{(8,),(16,),(32,)\} \times \text{learning_rate_init} \in \{0.001,0.01\}$, every epoch logged with `mlflow.log_metric(..., step=epoch)` to a local server on port 4000.

Best run: `mlp-32-lr0.001` at **0.9313** val accuracy. Capacity drives the ranking — 32 beats 16 beats 8 at *both* learning rates.

Overfitting: `train_loss` and `val_accuracy` move apart. `mlp-32-lr0.01` reaches the lowest training loss of the sweep (0.052) yet validates *below* `mlp-32-lr0.001` (0.923 vs 0.931), whose loss is $1.7\times$ higher. At 16 units, lr 0.01 cuts loss 0.151 $\rightarrow$ 0.102 and loses 1.5 points of val accuracy while the train/val gap widens 0.039 $\rightarrow$ 0.060.

Larger effect: architecture, by about $2\times$ — varying width moves val accuracy 0.0271 on average, varying learning rate only 0.0138.

Run ID:	7b08b3327b4463b3a3d37...	2780507b9047b0a034c5...	9c00008b734c046c70ee1a...	84367bf12d442320b177d1...	7a080430403026b777e1...	7c321232134403b067b4...
Run Name:	mlp-32-lr0.01	mlp-32-lr0.001	mlp-16-lr0.01	mlp-16-lr0.001	mlp-8-lr0.01	mlp-8-lr0.001
Start Time:	08/30/2026, 11:02:24 PM	08/30/2026, 11:02:06 PM	08/30/2026, 11:01:51 PM	08/30/2026, 11:01:36 PM	08/30/2026, 11:01:23 PM	08/30/2026, 11:01:10 PM
End Time:	08/30/2026, 11:02:39 PM	08/30/2026, 11:02:24 PM	08/30/2026, 11:02:05 PM	08/30/2026, 11:01:51 PM	08/30/2026, 11:01:36 PM	08/30/2026, 11:01:23 PM
Duration:	15.4s	18.4s	14.2s	15.2s	12.9s	13.0s
Parameters						
Show diff only						
hidden_layer_sizes	(32,)	(32,)	(16,)	(16,)	(8,)	(8,)
learning_rate_init	0.01	0.001	0.01	0.001	0.01	0.001
Metrics						
Show diff only						
test_epoch	9	30	10	26	3	30
test_val_accuracy	0.936	0.931	0.917	0.924	0.9	0.908
test_train_loss	0.052	0.088	0.102	0.151	0.185	0.269
test_val_accuracy	0.923	0.931	0.908	0.923	0.891	0.908
test_val_f1_macro	0.922	0.931	0.906	0.922	0.889	0.907
train_val_gap	0.058	0.048	0.06	0.039	0.048	0.016

MLflow run-comparison table for all six runs, with Show diff only enabled on both the parameter and metric panels.

Q3 — DVC Data Versioning & Rollback (repo root; tags v1, v2)

`dvc init`, then `data.zip` $\rightarrow$ 1800 jpgs $\rightarrow$ `file_list.csv` (1801 lines = 1800 rows + header) $\rightarrow$ `dvc add data file_list.csv`, pushed to an SSH remote: `dvc remote add -d ssh-remote ssh://samrudh010.17.64.87/research/.../dvcstore` followed by `dvc push -j 8` (key installed with `ssh-copy-id`; the keyfile lives in the `--local` config so no credential is committed). Tagged `v1`. Unzipping `new-labels.zip` gives 2800 jpgs and a 2801-line CSV; re-running `dvc add` rewrites only the md5 pointers, committed as “*data v2 with 2800 files*”, tagged `v2` and `dvc` pushed again. **Rollback**: `git checkout v1` rewinds the pointers alone — `dvc status` then reports both outputs modified — and `dvc checkout` swaps the workspace bytes back, restoring **1801 lines / 1800 files** exactly.

```
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ wc -l file_list.csv
881 file_list.csv
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ find data -type f -name "*.jpg" | wc -l
1800
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ git checkout v1
Note: switching to 'v1'.
HEAD is now at f2787bd data v1 added with 1800 files
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ dvc status
data.dvc:
  changed outs:
    modified:      data
  File list.csv.dvc:
    changed outs:
      modified:    file_list.csv
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ dvc checkout
Building workspace index [2.81k [00:00, 40.4kentry/s]
Comparing indexes [2.81k [00:00, 113kentry/s]
Applying changes [1.90 [00:00, 40.8kfile/s]
^
  file_list.csv
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ dvc status
data and pipelines are up to date.
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ wc -l file_list.csv
881 file_list.csv
(alops) samrudh@samrudh-HP-ProBook-440-14-Inch-G11-Notebook-PC:~/IITM/Courses/Sem 5/AI Operations Lab DA3408/Assignment-1/DA3408-Assignment-1$ find data -type f -name "*.jpg" | wc -l
1800
```

Rollback proof: 2801 lines / 2800 files $\rightarrow$ `git checkout v1` $\rightarrow$ `dvc checkout` $\rightarrow$ 1801 lines / 1800 files.

```
diff --git a/file_list.csv.dvc b/file_list.csv.dvc
index e4da9b1..94fe97e 100644
--- a/file_list.csv.dvc
+++ b/file_list.csv.dvc
@@ -1,5 +1,5 @@
-  outs:
-    md5: 105e0cb7df083267d765a5f5b77aae7
-    size: 54993
-+ md5: 9b9cc0f19067e06b2db12474c963d511
-+ size: 82995
-  hash: md5
-  path: file_list.csv
diff --git a/data.dvc b/data.dvc
index ee37909..0187709 100644
--- a/data.dvc
+++ b/data.dvc
@@ -1,6 +1,6 @@
-  outs:
-    md5: b0f4d5a70e55e08906d5f4aeaf43082e.dir
-    size: 41149004
-    nfiles: 1800
-+ md5: 21060888834f722046dc1c6fc04e649.dir
-+ size: 64128504
-+ nfiles: 2800
-  hash: md5
-  path: data
```

git diff of the .dvc pointers, v1 $\rightarrow$ v2: nfiles 1800 $\rightarrow$ 2800.

Q4 — End-to-End Reproducibility Drill (q4/question_4.ipynb; recording q4_video.mp4)

Partner A trained `MLPClassifier((128,64))` on full MNIST with `seed=42`, logging params, metrics, per-epoch curves and provenance tags (`git_commit`, `git_dirty`, `dataset_md5`, `dataset_dvc_md5`, versions), versioned the data with DVC to an S3 remote, committed `data.dvc` in the same commit as the code, and registered `mnist-mlp-q4` into **Staging**. Baseline: `sujal_run.json`, tolerance ± 0.005 .

Partner B used only `git clone`, `git checkout`, `conda env create -f environment.yml`, `dvc pull` (restores data/) and a re-run of the notebook against `mlflow server --backend-store-uri sqlite:///mlflow.db`.

metric	baseline	repro	delta	ok
accuracy	0.9751	0.9751	+0.0000	yes
f1_macro	0.9750	0.9750	+0.0000	yes

Verdict: REPRODUCED. `dataset_md5`, `seed` and `sklearn 1.9.0` match; only `git_commit` and the Python patch release (3.13.14 vs 3.13.15) differ, neither of which moved the metrics.

Metrics (0)		
Search metrics		
Metric	Value	Models
accuracy	0.9751428571428571	model
f1_macro	0.974967024690898	model
train_accuracy	0.9862142857142857	model
generalization_gap	0.02207142857142857	model
train_loss	0.00834995399538109	model
val_accuracy	0.966071428571429	
Parameters (0)		
Search parameters		
Parameter	Value	
hidden_layer_sizes	(128, 64)	
learning_rate_init	0.001	
batch_size	32	

Partner B's reproduction run: identical metrics, `repro_verdict: REPRODUCED`, `repro_delta_*: +0.000000`, registered as `mnist-mlp-q4 v3`.